DOI: 10.1111/tgis.12447

WILEY Transactions in GIS

RESEARCH ARTICLE

A dynamic human activity-driven model for mixed land use evaluation using social media data

Hanfa Xing[1,2] | Yuan Meng[2] | Yan Shi[3,4]

[1]School of Geography, South China Normal University, Guangzhou, China

[2]College of Geography and Environment, Shandong Normal University, Jinan, China

[3]Department of Geo-informatics, Central South University, Changsha, China

[4]Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan, China

Correspondence
Yuan Meng, College of Geography and Environment, Shandong Normal University, Jinan, Shandong, China.
Email: mengyuan@stu.sdnu.edu.cn
and
Yan Shi, Department of Geo-informatics, Central South University, Changsha, Hunan, China.
Email: csu_shiy@csu.edu.cn

Abstract

Massive amounts of social media data have enabled the evaluation of mixed land use from the human activities perspective. While most studies have focused on approaches to better measure mixed land use, the dynamic activity patterns used to distinguish land use types have seldom been discussed. In light of this, in this study, a mixed land use evaluation model is constructed, which is driven by dynamic human activities hidden in social media data. Specifically, temporal topics related to human activities are first extracted from Twitter messages. On this basis, these topics are clustered with successive measurements of both temporal and semantic similarities, and then, they are further transformed into dynamic features. In addition, the dynamic feature-based Shannon entropy is employed for mixed land use evaluation. For verification, the building classes from OpenStreetMap are referenced to investigate their correlations with dynamic features in mixed land uses. Finally, a strong correlation demonstrates the effectiveness and power of the proposed method in the evaluation of mixed land uses.

1 | INTRODUCTION

Mixed land use represents the functional integration of variable land use within different urban structures and socioeconomic features (Jacobs, 1961; Ma & Chen, 2013; Song & Knaap, 2004). The impacts of mixed land use on revealing neighborhood vibrancy (Yue et al., 2017), determining the commuting choices of residents (Cervero, 1996), establishing key principles for city planning (Grant, 2002), and discovering health-related outcomes (Brown et al., 2009) have been widely explored in recent studies. In this case, large amounts of methodological analyses of mixed land use measurements are proposed to better understand the diversity of land uses (Abdullahi, Pradhan, Mansor, & Shariff, 2015; Seasons, 2014; Song, Merlin, & Rodriguez, 2013). Most of the current research directly adopted static land use classes extracted from existing nomenclatures (Gehrke & Clifton, 2018), which were laboriously obtained through either remote sensing images or direct observations (Heiden et al., 2012), to unveil physical urban structures. However, the dynamic human activity information will be lost when extracting mixed land use indicators, which are highly relevant to the functions of places (Gao, Janowicz, & Couclelis, 2017) and can have great influence on land use diversity.

The recent emergence of crowdsourced data has offered opportunities to portray human activities (Gong, Lin, & Duan, 2017; Liu et al., 2015; Liu, Wang, Xiao, & Gao, 2012; Yao, Li, et al., 2017), and provides a new perspective for understanding mixed land use. Among all types of crowdsourced data, social media data, which includes both temporal and semantic information, has received much more attention due to its wide usage and free acquisition. However, there exist several challenges in utilizing social media data for mixed land use evaluations. One problem is that when evaluating mixed land use, there is a lack of consideration for temporal change characteristics embedded in different land use types. In fact, Liu et al. (2015) have discussed the temporal issue of detecting land use using social media, indicating that changes in temporal patterns can help to reveal the regional evolution of land use distribution. However, social media has not yet been considered for measuring mixed land use. Another challenge is that the same urban structures do not always share similar human activity patterns (Hu, Yang, Li, & Gong, 2016; Liu et al., 2017), which include both temporal and semantic patterns generated from vast amounts of social media data, and this could lead to land use variation to a large extent. Faced with the aforementioned challenges and driven by dynamic human activities, in this study, a new model for evaluating mixed land use using social media data is developed.

The remainder of the article is organized as follows. Section 2 reviews related studies of land use analysis using social media data. Section 3 elaborates the proposed model of mixed land use evaluation using social media data. A brief description of the study area and experimental data are provided in Section 4. Section 5 demonstrates the performance of the proposed model through practical experiments using Twitter messages in Toronto. Finally, in Section 6, the conclusions and future work to address the limitations of this study are presented.

2 | LITERATURE REVIEW

Social media data has been utilized extensively to delineate human activities in land use analyses. Related applications include land use classification (Frias-Martinez & Frias-Martinez, 2014), urban functional region detection (Chen et al., 2017; Niu et al., 2017), population estimation (Yao, Liu, et al., 2017), and mixed land use evaluation (Liu et al., 2018; Yue et al., 2017). Compared with other sources of data [such as points of interest (POI)], social media data can reveal regular patterns of dynamic human activities at the individual scale, instead of simply capturing the static distribution of land use forms. Furthermore, unlike metro smartcard data and taxi trajectories, which are usually generated for a unique purpose during specific time periods (Gong et al., 2017; Gong, Liu, Wu, & Liu, 2016; Liu et al., 2015), social media data can be updated constantly, whenever and wherever users intend to post their activities. The above characteristics combined allow researchers to better explore land use patterns from social media data.

The identification of land use types relies heavily on socioeconomic properties that can be derived from human activities. Two types of activities are usually considered in the current studies, which include temporal changes and semantic information generated by citizens. Temporal changes usually refer to the variety of human activities that occur during a typical time period. To explore the impact of temporal changes on land use, Ríos and Muñoz

(2017) proposed a novel method to detect land use patterns based on temporal variations in cell phone data. Frias-Martinez and Frias-Martinez (2014) investigated Twitter check-in changes for both weekdays and weekends and explored the corresponding land use based on the temporal patterns. Regarding the semantic information, Yuan, Zheng, and Xie (2012) built a framework known as DRoF for regional land use detection using human mobility and textual information. A similar study was proposed by Yao, Li, et al. (2017), in which the Word2Vec model was adopted to detect the spatial distribution of urban land use. Relationships between human activities and land use have been addressed and evaluated in the above mentioned studies. However, they have focused solely on the identification of specific land use types and have not considered the degree of mixture among different land uses.

To fill this gap, a variety of approaches have been proposed to apply human activities to land use mixture evaluations. For example, Yue et al. (2017) illustrated the relationship between human activity vibrancy and mixed land use, in which the diversity of POIs was considered to be a significant indicator of mixed land use. It also provided a new perspective on land use mixture measurements. Based on this hypothesis, Liu et al. (2018) integrated multiple types of crowdsourced data including social media data, taxi trajectories, and POIs to characterize mixed land use at the building level. In particular, a probabilistic model was built by extracting the characteristics of citizen mobility and location information, which included pick-up and drop-off locations, arrival times and trip lengths from taxi trajectories, and geo-locations in POIs. Additionally, data from OpenStreetMap (OSM) generated by users were also utilized in the mixed land use analysis (Gervasoni, Bosch, Fenet, & Sturm, 2016). Land use mixtures were calculated by an entropy index based on land use densities calculated by the kernel density estimation.

Although substantial progress has been made in utilizing human activity for measuring mixed land use, several researches have reviewed the limitations of social media data, including the complex process for extracting and leveraging data, as well as strategies for linking unorganized textual information to actual behaviors (Lansley & Longley, 2016; Liu et al., 2015). With these problems in mind, one major hypothesis is that social media data are capable of capturing regular and general patterns of human activities instead of typical events, and thus, this offers the potential to measure the diversity of land uses. In addition, to organize textual information and reduce data processing complexity, representative topics with specific keywords can be extracted from the noisy social media data and matched with different land uses. However, challenges are not limited to these discussions.

In fact, the impact of temporal and semantic changes in human activities on mixed land use has been ignored. On the one hand, land use mixture evaluations should be more sophisticated and incorporate more temporal characteristics to better capture the diversity of land use. This has been challenging because human activities generate semantic information to reflect different land uses and dynamically change with a variety of temporal patterns (García-Palomares, Salas-Olmedo, Moya-Gómez, Condeço-Melhorado, & Gutiérrez, 2018). Thus, the incorporation of temporal changes in human activities is required to comprehensively interpret the land use mixture. On the other hand, due to the diversity and uncertainty of human activities in different land use regions, it is essential to propose effective approaches to measure activity changes behind land use. These include measuring the similarities of both temporal and semantic changes. Based on the above two stated issues, a dynamic human activity-driven model is proposed for land use mixture evaluation.

3 | METHODOLOGY

This section will provide a detailed explanation of the mixed land use evaluation driven by dynamic human activities, which is based on social media data. As this type of data usually contains large amounts of unorganized textual information, topics containing specific keywords can be extracted to differentiate land use types. For example, when one topic includes high-probability keyworks such as “shopping,” “high price,” “clothes,” and “quality,” this topic may refer to commercial land use. Furthermore, temporal characteristics may vary among different topics. For example, one may hardly extract any residential-related topics during typical working periods.

Based on this hypothesis, a dynamic human activity-driven model is proposed (Figure 1), which includes three processes: temporal topics extraction with topic modeling in Section 3.2, dynamic features clustering from

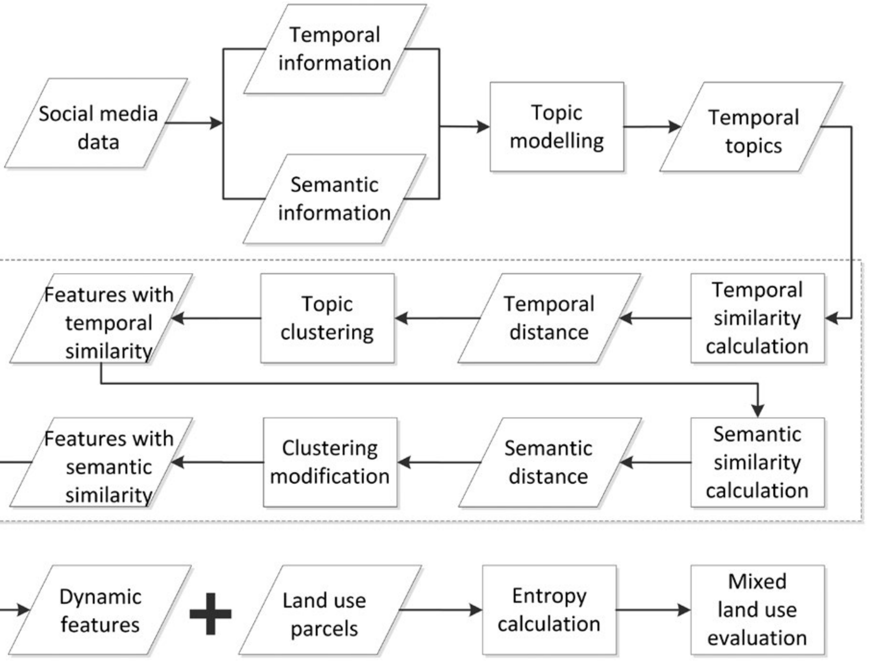

FIGURE 1 Workflow of the dynamic model for evaluating mixed land use

temporal topics in Section 3.3, and mixed land use evaluation using dynamic features in Section 3.4. To integrate dynamic human activities into mixed land use evaluation, potential topics with specific temporal distributions are calculated using topic modeling. Those topics with similar temporal and semantic distributions, which indicate topics containing a variety of time series and textual information, are clustered and transformed into distinct dynamic features. Eventually, the dynamic features are applied to land use parcels as indicators for the mixed land use evaluation using the Shannon entropy.

3.1 | Notation and terminology

For convenience, some basic notations and terminology in the proposed model are elaborated as follows.

3.1.1 | Latent Dirichlet allocation topic model

The Latent Dirichlet allocation (LDA) model has commonly been used for textual information analysis in existing research (Lansley & Longley, 2016; Xing, Meng, Hou, Song, & Xu, 2017; Yuan et al., 2012). Generally, one document d_i consists of different words w, and the aim of the LDA topic model is to extract topics that contain words with similar meanings. Moreover, two probability types can be calculated, including the probability Θ for one topic in document d_i and the probability ϕ for one word w in a specific topic. Based on this model, temporal characteristics will be introduced in this study to better delineate human activities in Section 3.2.

3.1.2 | Hierarchical clustering

This is a clustering method that seeks to build a hierarchy of clusters, which mainly include single-linkage and complete-linkage clustering. Specifically, the complete-linkage strategy is utilized for clustering due to its better performance (Hubert, 1974; Williams, Lance, Dale, & Clifford, 1971). To determine which data should be clustered, similarities between datasets need to be captured. In this study, similarities related to both temporal and semantic patterns will be calculated to extract dynamic features in Section 3.3.

3.1.3 | Shannon entropy

This is a measurement of unpredictability of the state, which has been used widely in computing the diversity of the datasets. In this study, the aim is the application of the Shannon entropy for evaluation of mixed land use, which is represented by dynamic features in Section 3.4.

3.2 | Temporal topics extraction with topic modeling

To sufficiently extract topics from social media data, the LDA (Blei, Ng, & Jordan, 2003) topic model was widely used in previous studies. However, these studies rarely considered the variable temporal distributions of different topics, and thus, they could not reflect dynamic human activities. Therefore, an LDA topic model is employed to extract temporal topics involving both semantic and temporal information.

Given a study area, all the semantic information limited to this area is considered to be document ***D***. The attached temporal information in ***D*** is split into 48 hr:

$$D=\begin{cases}\left(d_1,d_2,d_3\dots\dots d_t\dots\dots d_{24}\right), & t\in[1,24]\\ \left(d_{25},d_{26}\dots\dots d_t\dots\dots d_{48}\right), & t\in[25,48]\end{cases} \tag{1}$$

with the first 24 hr belonging to the mean values of weekdays, and the second 24 hr summarizing weekends and holidays based on the conclusion that human activities show different patterns between weekdays and weekends/holidays (Frias-Martinez & Frias-Martinez, 2014; Zhu, Wang, Wu, & Liu, 2017). This temporality based document *D* is established as the input for the LDA topic modeling:

$$p(\theta,z,w|\alpha,\beta)=p(\theta|\alpha)\prod_{n=1}^{N}p\left(z_n|\theta\right)p\left(w_n|z_n,\beta\right) \tag{2}$$

where *z* refers to the topics calculated by topic modeling, *w* represents the words in social media data, *n* is the number of words, and α,β are hyperparameters in LDA, in which $\alpha=50/k$, $\beta=0.1$ (Griffiths & Steyvers, 2004). *k* represents the number of topics selected by a *perplexity* algorithm (Blei et al., 2003). Θ refers to topic distributions in document ***D*** and can be represented by the following matrix:

$$\Theta=\begin{array}{cccccccc} & z_1 & z_2 & z_3 & \dots & z_i & \dots & z_k\\ d_1 & P_{1,1} & P_{1,2} & P_{1,3} & \dots & P_{1,i} & \dots & P_{1,k}\\ d_2 & P_{2,1} & P_{2,2} & P_{2,3} & \dots & P_{2,i} & \dots & P_{2,k}\\ d_3 & P_{3,1} & P_{3,2} & P_{3,3} & \dots & P_{3,i} & \dots & P_{3,k}\\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots\\ d_t & P_{t,1} & P_{t,2} & P_{t,3} & \dots & P_{t,i} & \dots & P_{t,k}\\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots\\ d_{48} & P_{48,1} & P_{48,2} & P_{48,3} & \dots & P_{48,i} & \dots & P_{48,k}\end{array} \tag{3}$$

where each row of Θ represents the distribution of topics during one time interval. Each column indicates the distribution of a specific topic in 48 time intervals, denoted as $\Theta_z=\left(P^z_1,P^z_2,P^z_3\dots P^z_{48}\right)$.

The temporal distributions of each topic z are considered to be its temporal changes during the 48 hr period. In addition, each value in the temporal distribution of each topic refers to the probability that this topic belongs to the corresponding time interval. To facilitate the evaluation of temporal changes among different topics, the temporal distributions of topics Θ_z need to be normalized:

$$\Theta_z = \frac{P_t^z - min(\Theta_z)}{max\left(\Theta_z\right) - min(\Theta_z)}, \quad t \in [1,48] \tag{4}$$

where $max\left(\Theta_z\right)$ and $min(\Theta_z)$ indicate the maximum and minimum values in the temporal distribution of topic z, respectively.

Recent studies have shown that the number of topics determined by the *perplexity* algorithm do not correlate strongly with the actuality (Chang, Gerrish, Wang, Boyd-Graber, & Blei, 2009). In other words, separate topics may indicate similar human activities from a semantic perspective. Additionally, temporal changes in different topics can be variable or similar, regardless of whether they share the same words or not. Therefore, targeted solutions are proposed in Section 3.3 to obtain the effective dynamic features for further evaluation of mixed land use.

3.3 | Dynamic features clustering from temporal topics

Since the extracted temporal topics may contain similar semantic and temporal characteristics, both semantic and temporal similarity will be considered to perform a hierarchical clustering to extract dynamic features, and will further be considered as indicators to evaluate mixed land use. In other words, while temporal topics represent the extracted semantic information with temporal patterns, dynamic features refer to the grouped topics in which the features contain different semantic and temporal characteristics from each other.

The process of dynamic feature clustering can be concluded in the following steps. The first step is to calculate the temporal distance between two temporal topics. In the second step, the dynamic features are grouped by similar temporal distance using a hierarchical clustering method. In particular, the similarities within each dynamic feature are determined by the number of clusters, which is evaluated using a Davies–Bouldin (DB) index (Davies & Bouldin, 1979). In the third step, the dynamic features which contain more than one temporal topic are further decomposed by calculating semantic distance and clustered according to the first two steps.

3.3.1 | Dynamic features clustering with temporal similarity

Regarding the measurement of temporal distance, dynamic time warping and edit distance on real sequences have been used widely in the related research (Chen et al., 2017; Chen, Özsu, & Oria, 2005; Tormene, Giorgino, Quaglini, & Stefanelli, 2009). Recent research by Serra and Arcos (2014) revealed the potential of the commonly unconsidered measure, time-warped edit distance (TWED) (Marteau, 2009), which consistently outperformed all the above measures. TWED comprises both a mismatch penalty and a stiffness parameter to control the elasticity of temporal similarity. In this case, this study applies TWED to calculate temporal distances between every pair of temporal topics.

Let Θ_{z_1}, Θ_{z_2} be the temporal distributions of two topics, where $\Theta_{z_1} = \left(P_1^{z_1}, P_2^{z_1}, P_3^{z_1} \ldots\ldots P_i^{z_1} \ldots\ldots P_{48}^{z_1}\right)$ and $\Theta_{z_2} = \left(P_1^{z_2}, P_2^{z_2}, P_3^{z_2} \ldots\ldots P_j^{z_2} \ldots\ldots P_{48}^{z_2}\right)$. The temporal distance between Θ_{z_1} and Θ_{z_2} is calculated as follows:

$$d_{TWED}\left(\Theta_{z_1}, \Theta_{z_2}\right) = D_{i,j} = min\left\{D_{i,j} + \Gamma_{\Theta_{z_1}\Theta_{z_2}}, D_{i-1,j} + \Gamma_{\Theta_{z_1}}, D_{i,j-1} + \Gamma_{\Theta_{z_2}}\right\} \tag{5}$$

where

$$\Gamma_{\Theta_{z_1}\Theta_{z_2}} = f\left(P_i^{z_1}, P_j^{z_2}\right) + f\left(P_{i-1}^{z_1}, P_{j-1}^{z_2}\right) + 2v\,|i-j|$$

$$\Gamma_{\Theta_{z_1}} = f\left(P_i^{z_1}, P_{i-1}^{z_1}\right) + v + \lambda \tag{6}$$

$$\Gamma_{\Theta_{z_2}} = f\left(P_j^{z_2}, P_{j-1}^{z_2}\right) + v + \lambda$$

Here, $f\left(P_i^{z_1}, P_j^{z_2}\right) = \left|P_i^{z_1} - P_j^{z_2}\right|$. λ and v are parameters of the mismatch penalty and stiffness parameter, respectively. In particular, mismatching indicates the inappropriate matching of two time series in temporal topics based on their similarity, and the mismatch penalty is utilized to penalize the matching of time series that are not similar enough to favor matching. For the selection of parameters λ and v, the ranges $v = \left[10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\right]$ and $\lambda = [0, 0.25, 0.5, 0.75, 1]$ were suggested in previous research (Serra & Arcos, 2014). In this study, the strategy of grid search is employed to measure the temporal distances between different temporal topics with all 25 (5 × 5) combinations of λ and v. In addition, the first nearest neighbor classifier on labeled temporal topics is utilized to evaluate the temporal distance measurement accuracy by grid search in terms of classification errors (Wang et al., 2013). The parameter combination with the smallest classification error will be chosen to determine the temporal distance.

Based on the temporal distance, the number of clusters should be determined to evaluate the similarity and obtain effective dynamic features. Here, a range of cluster numbers belonging to $[1,10]$ are utilized separately in hierarchical clustering, and the corresponding clustering results are evaluated through the DB index. The DB index aims to validate how well the clustering has been performed with a specific cluster number, where a lower DB value represents a better clustering result. In particular, the DB index is defined as:

$$DB = \frac{1}{N} \sum_{i=1}^{N} \max_{j \neq i} \left(\frac{S_i + S_j}{d_{TWED}\left(\Theta_{c1}, \Theta_{c2}\right)} \right) \tag{7}$$

where N refers to the number of clusters, S_i and S_j represent the standard deviations of the temporal distribution of topics in each cluster, and $d_{TWED}\left(\Theta_{c1}, \Theta_{c2}\right)$ indicates the temporal distance between cluster centroids Θ_{c1} and Θ_{c2}. The N with the minimum DB value will be determined as the proper number of clusters.

After dynamic features are extracted using hierarchical clustering with temporal similarity, further analysis is required to modify the dynamic features by considering the semantic similarity.

3.3.2 | Modifying dynamic features with semantic similarity

In regard to the clustered features with temporal similarity, the semantic similarity between different topics is further considered to modify these dynamic features. Previous studies usually measured the semantic similarity by calculating the frequency of particular words in each document. However, different words may express similar semantic information, which leads to unreliability of the word frequency calculation. Therefore, this study considers the potential distribution of words in each topic as the indicator of semantic similarity.

Specifically, the word distribution is generated from Equation (2), which is denoted as $\phi_z = \left(P_1^z, P_2^z, P_3^z \ldots\ldots P_i^z \ldots\ldots P_n^z\right)$, where n is the number of words in topic z. Then, the cosine distance is utilized to measure the semantic similarity between the distributions of words in two topics, ϕ_{z_1} and ϕ_{z_2}. As shown in Equation (8), potential distributions of the same word in topics z_1 and z_2 are first multiplied, which is followed by summarizing the multiplied distributions of all words. Then, the two square roots of the values are multiplied, and the values are calculated from the potential distributions of each word in two topics, which are squared and then summarized. The multiplied value in the first step is divided by the results obtained in the second step, which is considered to be the semantic similarity between the distributions of words for two topics:

$$similarity(\phi_{z_1},\phi_{z_2}) = \frac{\sum_{i=1}^{n} P_i^{z_1} P_i^{z_2}}{\sqrt{\sum_{i=1}^{n} {P_i^{z_1}}^2}\sqrt{\sum_{i=1}^{n} {P_i^{z_2}}^2}} \tag{8}$$

where n is the total number of words in each topic. Here, it should be noticed that the semantic similarity ranges from 0 to 1, in which a higher value represents a smaller distance between two topics. Thus, the semantic similarity can be transformed into the semantic distance:

$$distance\left(\phi_{z_1},\phi_{z_2}\right) = \frac{1}{similarity(\phi_{z_1},\phi_{z_2})} \tag{9}$$

The semantic distances between two topics with similar temporal distributions are further applied to perform the hierarchical clustering with the means from the DB index to determine the cluster number N. The modified dynamic features with both semantic and temporal similarities will be utilized for the evaluation of mixed land use.

3.4 | Mixed land use evaluation using dynamic features

Since dynamic features have been clustered based on both temporal and semantic similarity, these features contain sufficient temporal and semantic information that can effectively reflect human activities. Thus, dynamic features are considered to be significant indicators for evaluating mixed land use. The first step is utilizing land use parcels as the basic unit for the evaluation. In regard to the distributions of topics in each parcel, topic modeling with calculated temporal topics in Section 3.2 is then utilized considering land use units as document $\boldsymbol{D}$, where $D = (d_{p_1}, d_{p_2}, d_{p_3} \ldots d_{p_i} \ldots d_{p_n})$, with n the number of land use parcels. In addition, the distributions of topics in each parcel Θ_p is represented as follows:

$$\Theta_p = \begin{array}{cccccccc} & z_1 & z_2 & z_3 & \ldots & z_i & \ldots & z_k \\ p_1 & P_{1,z_1} & P_{1,z_2} & P_{1,z_3} & \ldots & P_{1,z_i} & \ldots & P_{1,z_k} \\ p_2 & P_{2,z_1} & P_{2,z_2} & P_{2,z_3} & \ldots & P_{2,z_i} & \ldots & P_{2,z_k} \\ p_3 & P_{3,z_1} & P_{3,z_2} & P_{3,z_3} & \ldots & P_{3,z_i} & \ldots & P_{3,z_k} \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ p_i & P_{i,z_1} & P_{i,z_2} & P_{i,z_3} & \ldots & P_{i,z_i} & \ldots & P_{i,z_k} \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ p_n & P_{n,z_1} & P_{n,z_2} & P_{n,z_3} & \ldots & P_{n,z_i} & \ldots & P_{n,z_k} \end{array} \tag{10}$$

In each land use parcel p_i, the dynamic features can be calculated as follows:

$$f_c^{p_i} = \sum_{m=1}^{M} P_{i,z_m} \tag{11}$$

where $f_c^{p_i}$ indicates the distribution of the cth dynamic feature located in land use parcel p_i. P_{z_m} refers to the distribution of topic z_m in this parcel, and M represents the number of topics in this dynamic feature. Therefore, the set of dynamic features F_{p_i} in parcel p_i can be denoted as $F_{p_i} = \left(f_1^{p_i}, f_2^{p_i}, f_3^{p_i} \ldots \ldots f_c^{p_i} \ldots \ldots f_C^{p_i}\right)$, where C is the number of dynamic features. On the basis of these dynamic features, the Shannon entropy is then adopted to evaluate mixed land use:

$$H(\boldsymbol{F}_{pi}) = -\sum_{i=1}^{c} f_i^{pi} \ln(f_i^{pi}) \tag{12}$$

where $H\left(F_{p_i}\right)$ indicates the mixed land use indices in parcel p_i, with higher and lower values of $H\left(F_{p_i}\right)$ representing more complex and simplified land use mixtures, respectively.

4 | STUDY AREA AND DATASETS

In our experiments, the downtown of Toronto, Canada is selected as the study area. It is generally known that this area is the most densely populated part of Toronto, and thus, Twitter messages are widely generated and spread by human activities, which are available and able to be applied to the mixed land use evaluation.

In the study area, Twitter messages with geo-locations from April 1, 2014 to April 1, 2015 are selected. Despite not all the Twitter messages being geotagged, they can still reveal the significant influence of human activity trends. In fact, among the total number of 17,403,472 Twitter messages in Toronto, 13,534,650 of them contain detailed geo-locations and occupy approximately 77.8% of the total messages, which indicates that no significant biases are shown in the geotagged data.

Given that the focus of this article is on measuring mixed land use from temporal and semantic perspectives, semantic information, published location, and date and time from each Twitter message are extracted for use in this study. These Twitter messages require a cleaning process to keep the effective data for measuring mixed land use and filter noise. Specifically, Twitter messages with fewer than three words are excluded from the experimental data. The remaining semantic information from Twitter messages is then processed by removing stop words (such as "the," "is," "at," "which," and "on"), URLs, and non-English words. Furthermore, a word frequency is calculated, and only words appearing over 15 times in Twitter messages are selected for the following semantic analysis. As a result, a total of 2,883,571 Twitter messages remained in the study area. To evaluate mixed land use considering dynamic human activities, these Twitter messages are divided into two parts according to their published date. In this way, 1,946,523 and 937,048 Twitter messages for weekdays and weekends/holidays, respectively, are extracted.

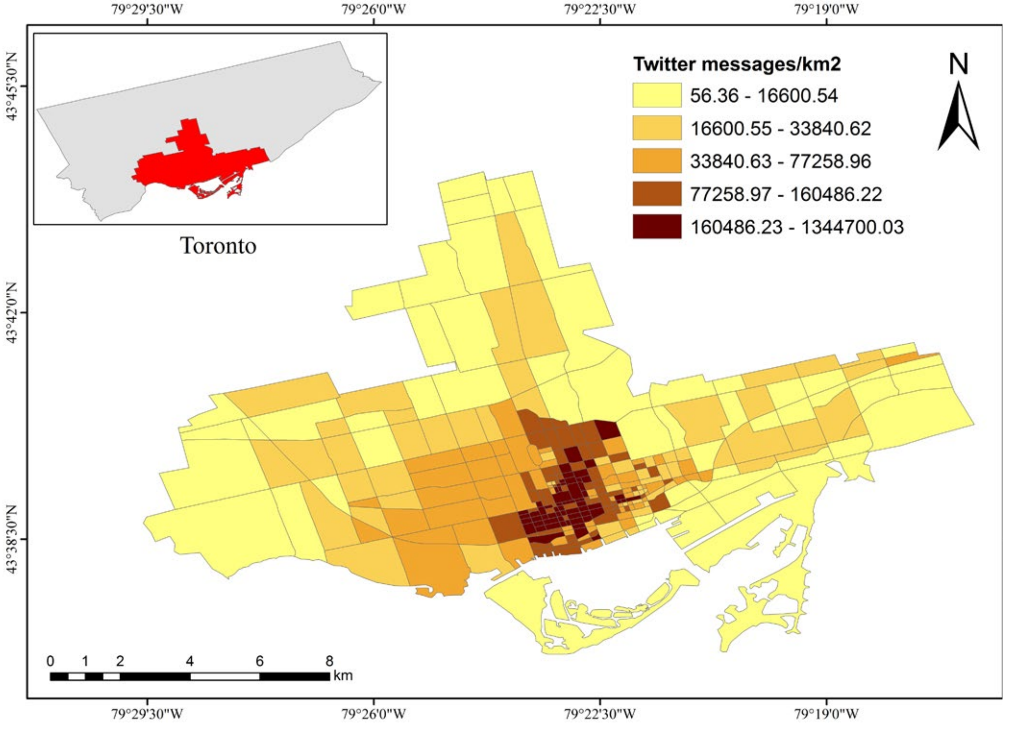

FIGURE 2 Land use parcels split by road networks and Twitter message density in the study area

Additionally, utilizing the road network provided by OSM, the main roads in the study area are both extracted and assessed based on high spatial resolution images from Google Earth. As a result, 880 roads are obtained, which are split by their intersected nodes. As these roads have been compared with high spatial resolution images during the extraction process, the quality of the roads is ensured. The extracted roads are utilized to split the overall study area into a series of land use parcels. A total of 305 parcels are obtained to represent the basic land use for the following mixture evaluation. When overlain with the spatial distribution of Twitter messages, the parcel without any messages published is excluded from the experiments. Figure 2 shows the Twitter message density for each land use parcel, in which relatively high-density data are located in the central area.

5 | RESULTS AND DISCUSSION

5.1 | Temporal topic extraction

The *perplexity* is first computed to determine the number of temporal topics in Twitter messages, where number $k \in [2, 20, 30, 40 \ldots \ldots 100]$. Results show that the *perplexity* value decreases dramatically as the k value increases gradually from 2 to 30, which is followed by a slight reduction of values from $k = 30$ to 100. A *perplexity* score indicates a much better performance, and a slight value reduction does not have a great impact on topic generation. As a result, $k = 30$ will be chosen as the proper number of temporal topics.

After exploring the 30 total temporal topics, the temporal distributions of topics in 48 time intervals are acquired by LDA topic modeling, and these are considered to be the temporal changes in topics. The distributions of each topic were normalized in the range of [0,1], and the results are shown in Figure 3, where [1, 24] and [25, 48] indicate the 24 hr on weekdays and weekends/ holidays, respectively. The light blue color indicates a high value in the topic distribution. A visual inspection shows that the peak value appears in different time intervals with respect to various topics. Some topic distributions, such as topic 11, reveal similar patterns on weekdays and

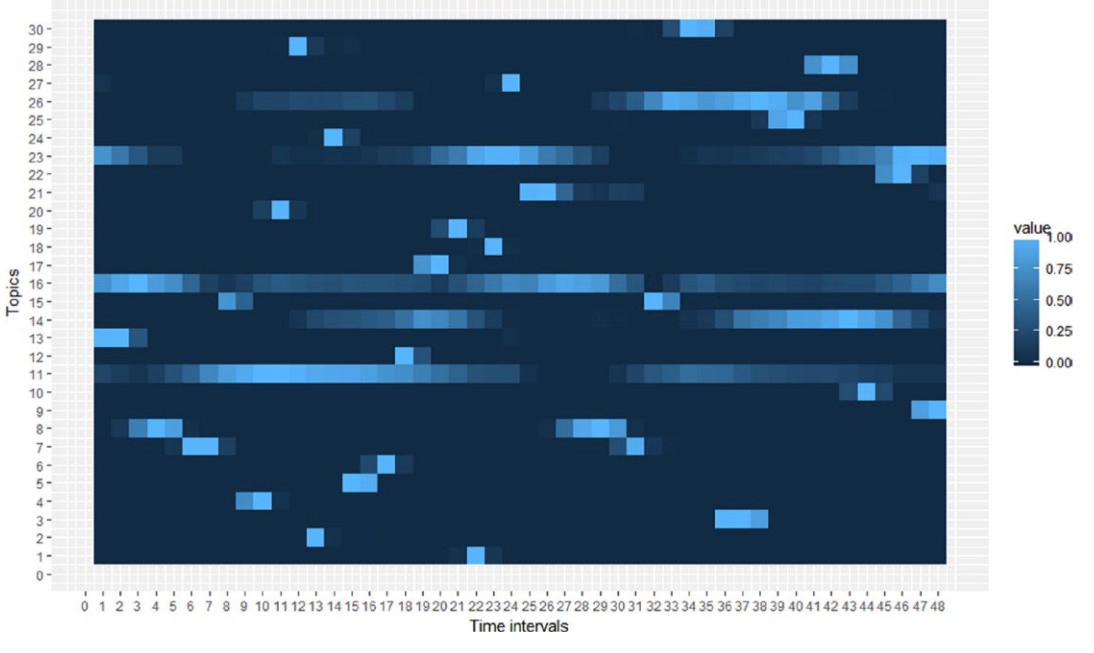

FIGURE 3 Temporal distributions of topics in 48 time intervals

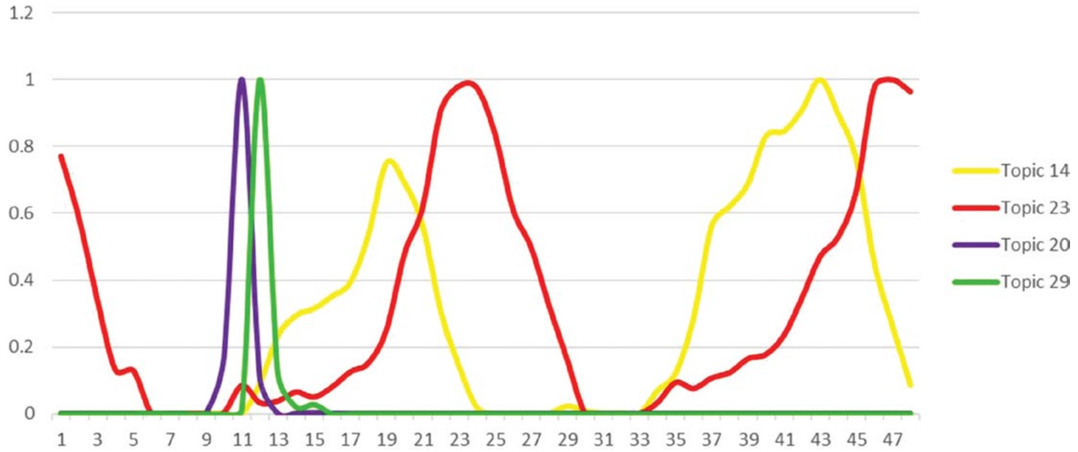

FIGURE 4 Temporal distributions of topics 14 and 23, as well as topics 20 and 29

weekends/holidays, which indicates similar dynamic human activities to some extent. Most of the topics reflect completely different dynamic patterns on weekdays and weekends/holidays. Using topic 5 as an example, the peak value is only present from approximately 13:00 to 17:00 on weekdays, which is probably caused by the high frequency of human activities during this period. Since different types of land use reflect the diversity of human activities, the temporal information within a given topic has the function of exploring mixed land use in a city.

In fact, the extracted topics with the selected number *k* could not sufficiently summarize those topics with similar temporal and semantic information. For example, Figure 4 compares the differences in temporal distributions between topics 14 and 23. Both topics experience two peaks in the afternoon and at night on weekdays and weekends/holidays, but there is a more delayed rush hour for topic 23. For the distribution of topic 23, a drastic decrease occurs in the morning on weekdays, compared with the static status in topic 14 during this period. However, Figure 5 reveals similar semantic patterns between topics 14 and 23, which both represent leisure and related nightlife. Meanwhile, topics 20 and 29 share similar temporal patterns, where the peak arises from 9:00 to 13:00 on weekdays. From the semantic perspective, the two topics present different semantic patterns, which are shown in Figure 5. While topic 20 indicates more about normal daily life, including “morning” and “breakfast,” topic 29 mainly indicates working and jobs. Therefore, it is necessary to sufficiently extract the dynamic features based on the similarities in semantic and temporal patterns.

5.2 | Dynamic features extraction with temporal and semantic clustering

With the obtained temporal topics, the dynamic features can be extracted by clustering the topics based on both temporal and semantic similarities.

5.2.1 | Dynamic features with temporal similarity

Figure 6 shows the temporal distances among the 30 obtained topics. While lower values indicate higher temporal similarity between two topics, high values with a light color represent two relatively independent topics with variable temporal information. A visual inspection highlights the significant distinction between topics 16, 23, and other topics, as the temporal similarity values between these two topics and others are mostly located in the range of [6,7]. The other topics mostly share relatively similar temporal patterns. For example, the temporal distance between topics 20 and 29 has the lowest-value interval [0,0.5].

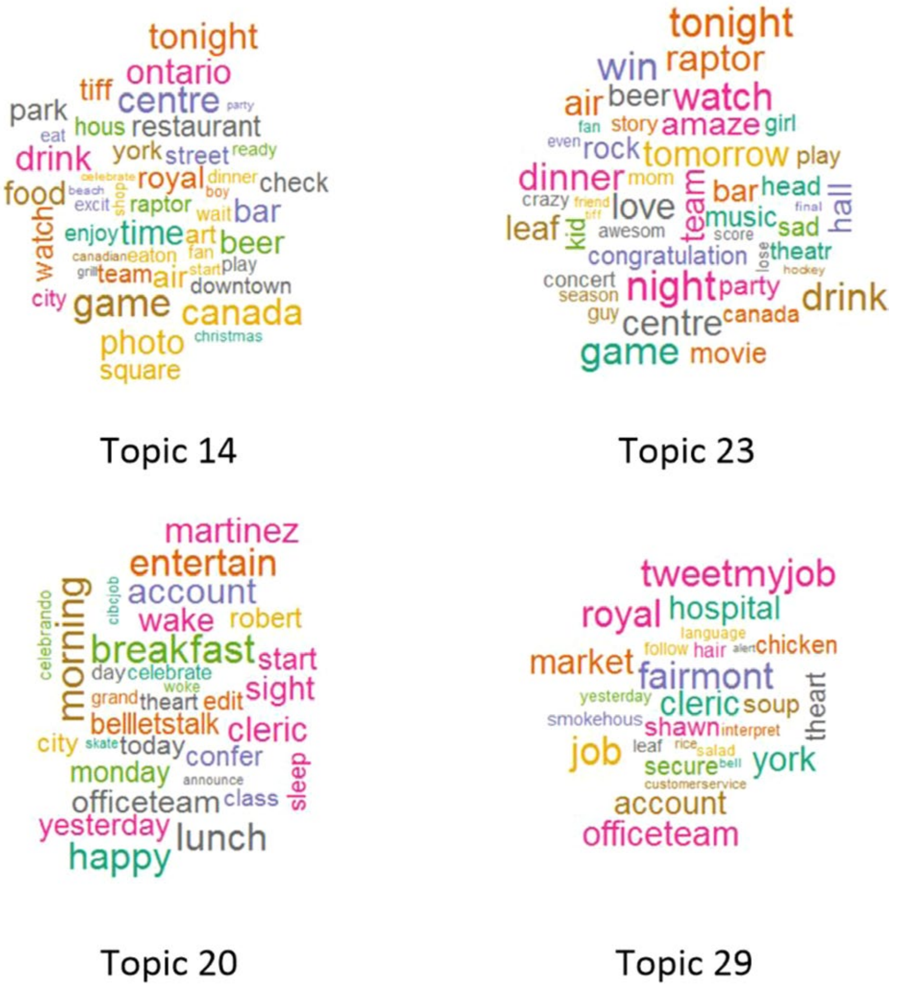

FIGURE 5 Word clouds of topics

The above temporal distances are applied to the hierarchical clustering process to group topics with temporal similarity. Based on the calculation of the DB index, the clustering results are shown in Figure 7. Topics 11, 13, 16, and 23 are separated into independent features. The cluster with topics 14 and 26, and the cluster including "other" topics, both share similar temporal information, respectively. The clustering results are consistent with the temporal information in Figure 4. As topics 20 and 29 share similar temporal patterns, they are clustered into the same dynamic features temporally in Figure 7. Meanwhile, topics 14 and 23 represent much different temporal changes, and they are separated as independent dynamic features according to the clustering results. However, the semantic similarity should be handled to remove the inconsistencies between topics in a similar temporal cluster. Therefore, 26 of the 30 topics, excluding topics 11, 13, 16, and 23, are used for further clustering analysis considering their semantic information.

5.2.2 | Modified dynamic features with semantic similarity

For the 26 topics with similar temporal information, the word distribution-based semantic similarity between two topics is further calculated and shown in Figure 8. In this matrix, higher values within the range of [0,1] indicate a larger dependency between two topics. Most of the topics show relatively low semantic similarity with others.

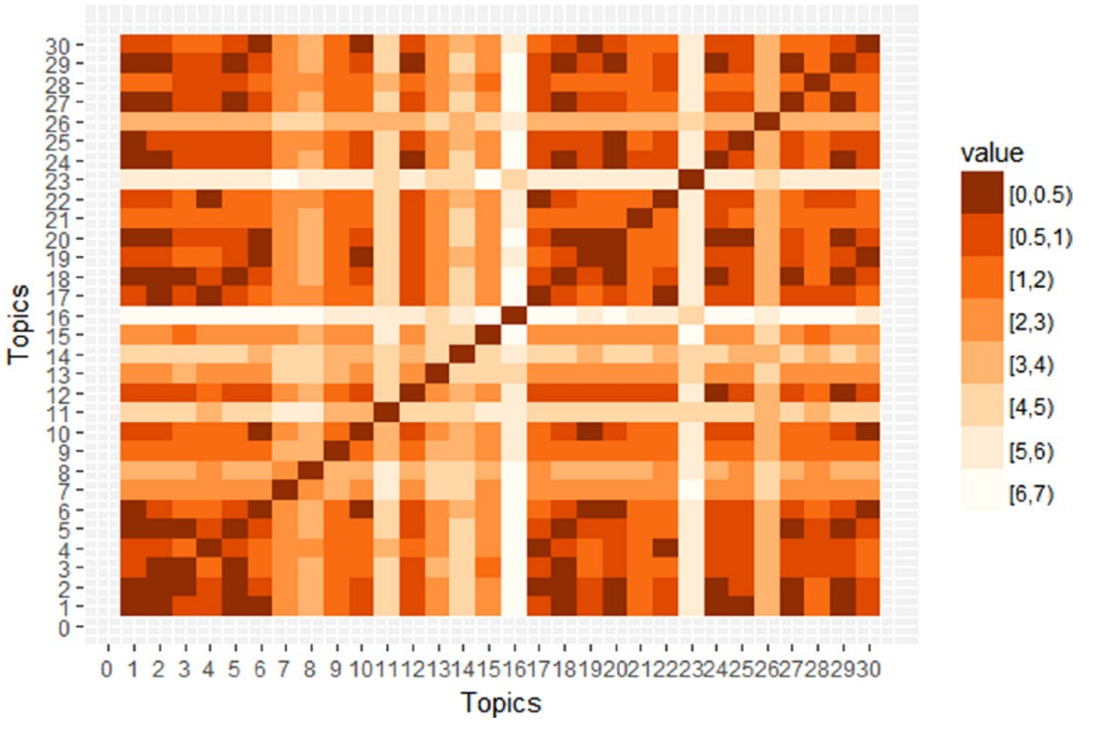

FIGURE 6 Temporal distances between temporal topics

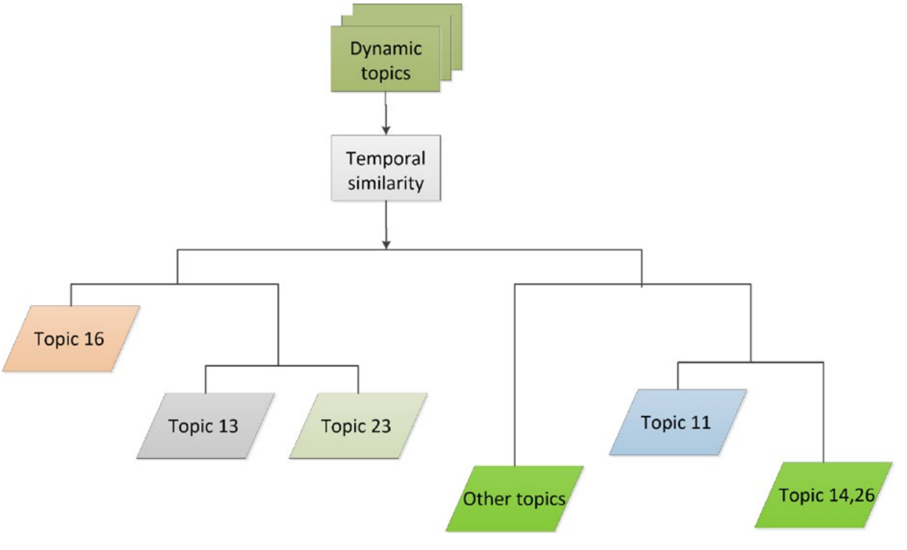

FIGURE 7 Clusters with temporal similarity

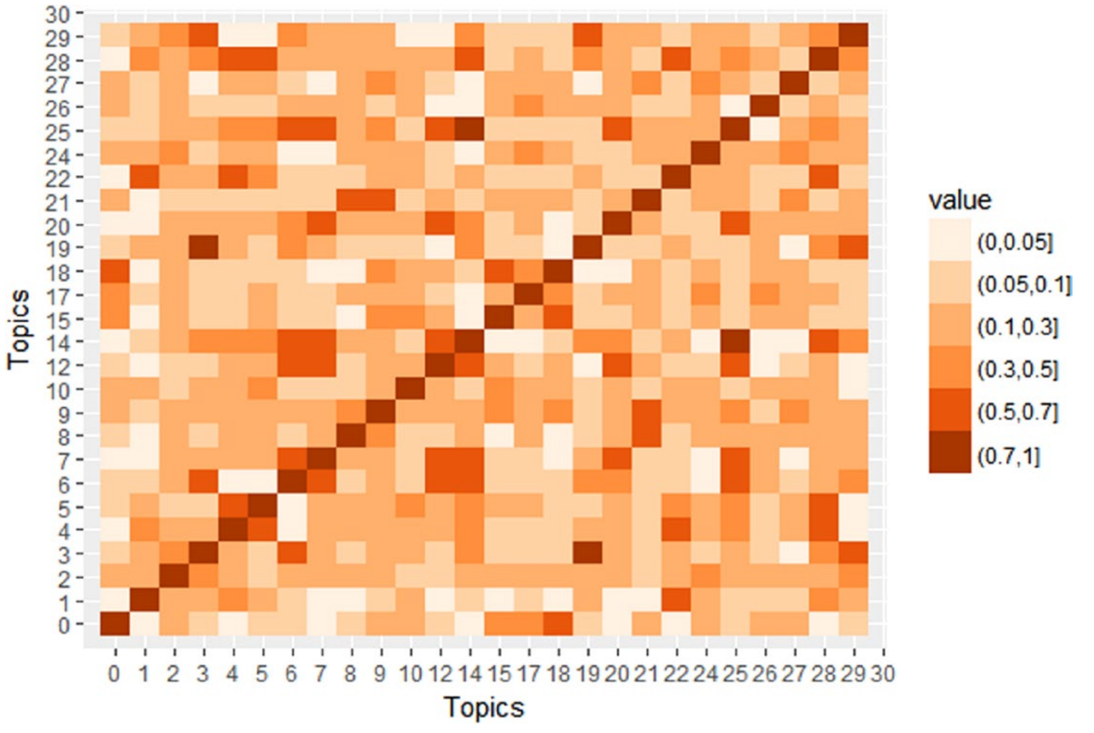

FIGURE 8 Semantic distance between temporal topics

Figure 9 exhibits the clustered topics based on semantic similarity. In the results, only topic 27 is separated as an independent feature, while the others all share similar semantic information with some other topics. As a result, all those topics belonging to the same cluster are considered one dynamic feature with both temporal and semantic similarity. Actually, the temporal and semantic patterns in Figures 4 and 5 prove the rationality of the extracted dynamic features. Based on the analyses in Section 5.1, topics 14 and 23 share similar semantic information but different temporal patterns, while topics 20 and 29 represent different semantic information disregarding the similar temporal changes. Thus, the above four topics are all grouped in the independent dynamic features, as shown in Figures 7 and 9.

5.3 | Mixed land use evaluation

By summarizing the clustering results shown in Figures 7 and 9, a total of 14 dynamic features are extracted from the 30 topics, as shown in Table 1. Among these, topics 11, 13, 16, 23, and 27 are independent from the others, and the remaining features usually contain two to four topics.

Then, these dynamic features with variable semantic and temporal patterns are further analyzed in the land use parcel units. Features 4 and 14, which contain the discussed topics 23 and 14, are taken as an example to make an elaboration. Although the two features both indicate leisure and nightlife from a semantic perspective, as shown in Figure 5, the discrepancy of temporal activities leads to different spatial distributions in the land use parcels. Specifically, feature 4 with topic 23 shows a more active pattern of nightlife than feature 14, which includes topic 14 during the period 00:00–7:00 on both weekdays and weekends/holidays. Figures 10a and b represent the spatial distributions of features 4 and 14, respectively, in each land use parcel. Feature 4 has high values and is widely distributed around the city center, but land use parcels in the southern part of the study area also contain higher values of feature 14.

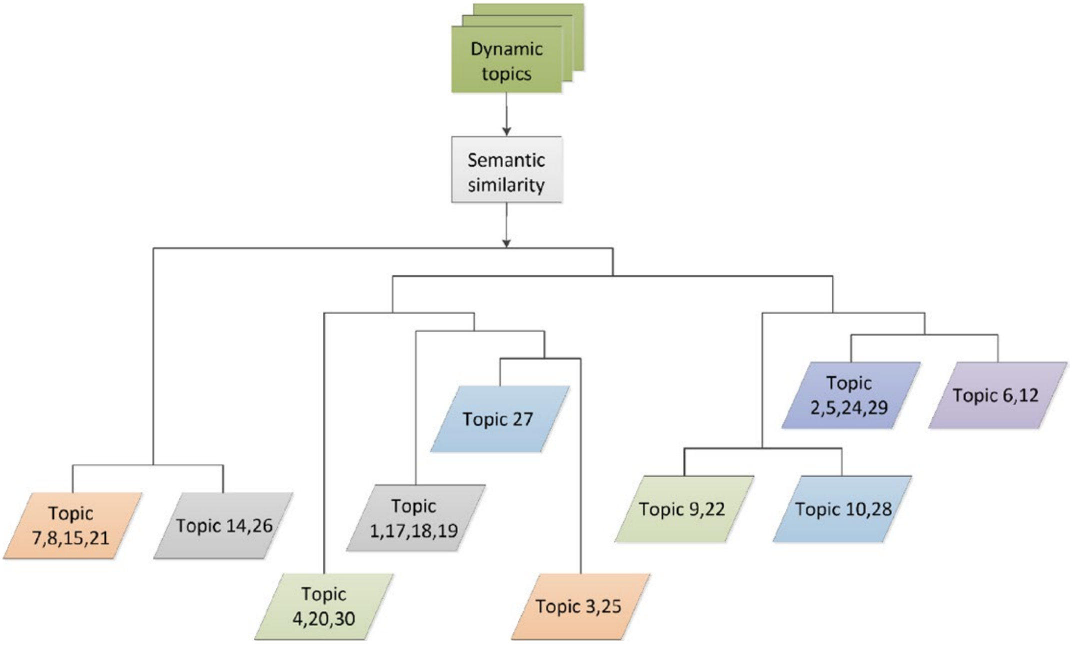

FIGURE 9 Clusters with semantic similarity

To quantitatively evaluate the mixture of land use, the 14 extracted features are embedded into the Shannon entropy to construct an indicator of land use diversity in each parcel, as shown in Figure 11. Much lower values of diversity are mostly located in the center of the study area, which indicates a simplified mixture of land use in these parcels. This reveals that areas with a relatively low value of mixed land use do not usually represent

TABLE 1 Dynamic features clustered from temporal topics

Dynamic features	Temporal topics
Feature 1	Topic 11
Feature 2	Topic 13
Feature 3	Topic 16
Feature 4	Topic 23
Feature 5	Topic 27
Feature 6	Topics 7, 8, 15, 21
Feature 7	Topics 4, 20, 30
Feature 8	Topics 1, 17, 18, 19
Feature 9	Topics 3, 25
Feature 10	Topics 9, 22
Feature 11	Topics 10, 28
Feature 12	Topics 2, 5, 24, 29
Feature 13	Topics 6, 12
Feature 14	Topics 14, 26

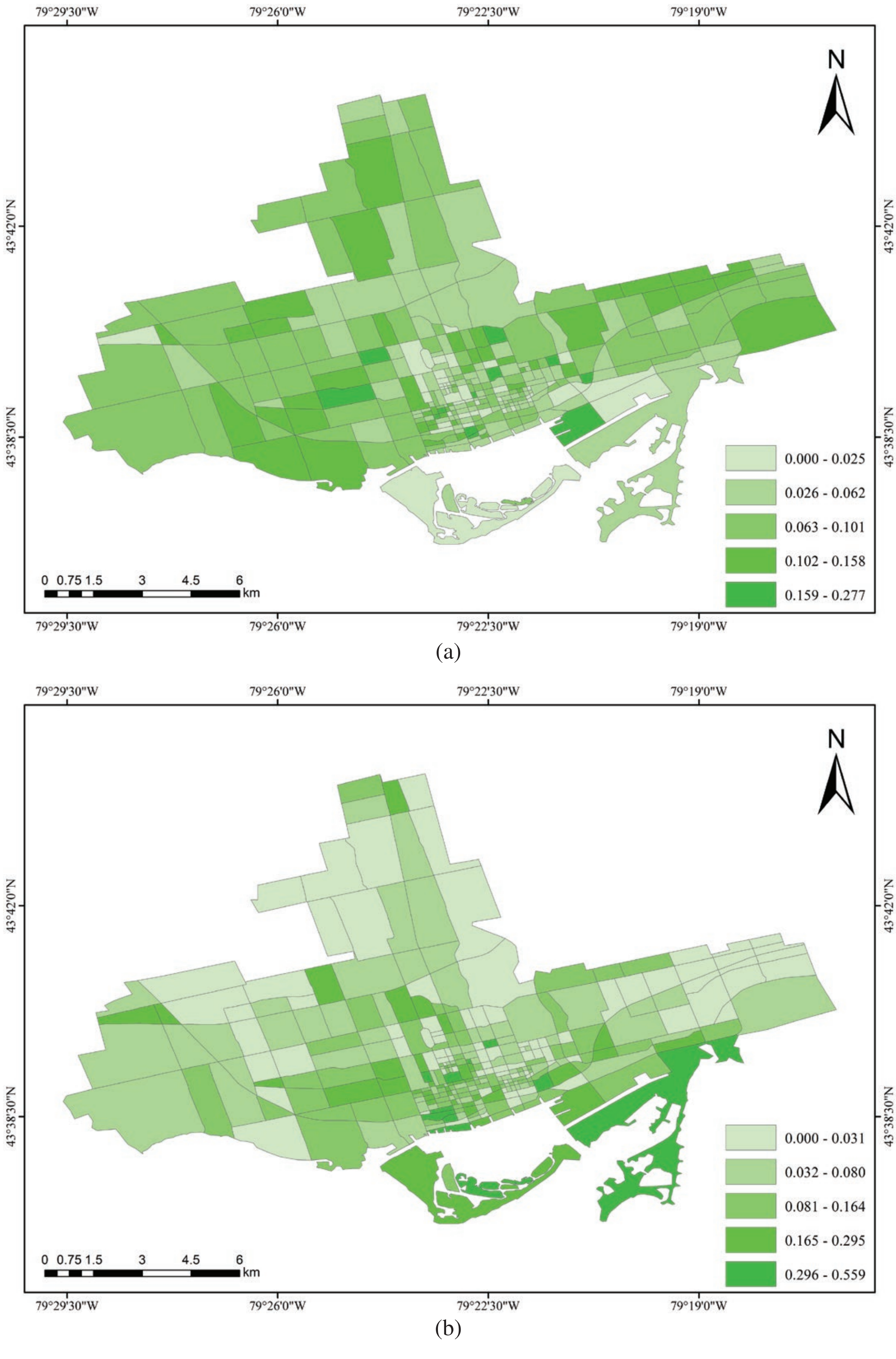

FIGURE 10 (a) Distributions of feature 4 in land use parcels ; and (b) Distributions of feature 14 in land use parcels

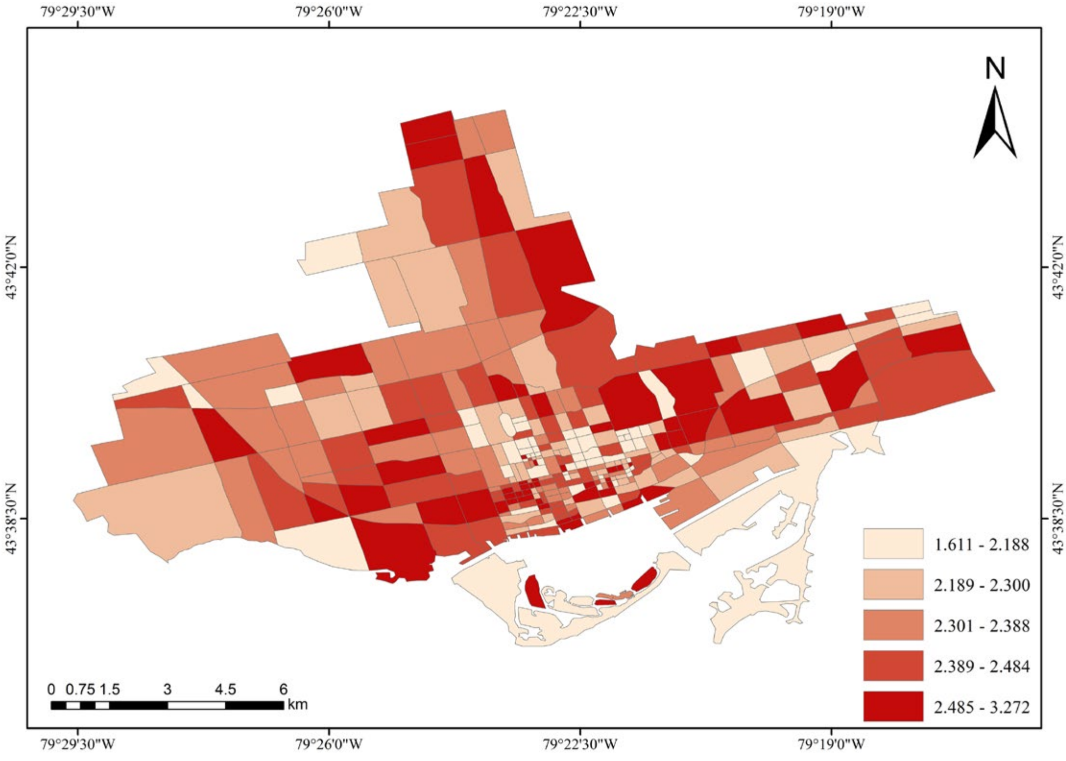

FIGURE 11 Mixed land use evaluation by Shannon entropy

undeveloped regions. In contrast, some areas that are far from the center have high values of diversity, which may be caused by the complexity of residential buildings and surrounding public facilities.

5.4 | Validation of mixed land use

To validate the evaluation of mixed land use with the proposed model, building-level blocks were utilized as the reference data to assess the land use diversity. Specifically, a total of 17,616 buildings tagged with certain types from OSM were extracted from the study area. Those buildings without land use types are excluded from the validation, and the remaining 9,227 buildings are classified based on the land use nomenclature proposed by

TABLE 2 Land use nomenclature adopted for validation

Code	Land use nomenclature
1100	Residential
1203	Isolated Commercial Establishments for Goods and/or Services
1204	Isolated Commercial Office Buildings
1206	Resorts, Hotels, Motels, and Related Facilities
1207	Educational Institutions
1208	Health Institutions
1212	Other Institutional
1810	Stadium, Theaters, Cultural Centers, and Zoos

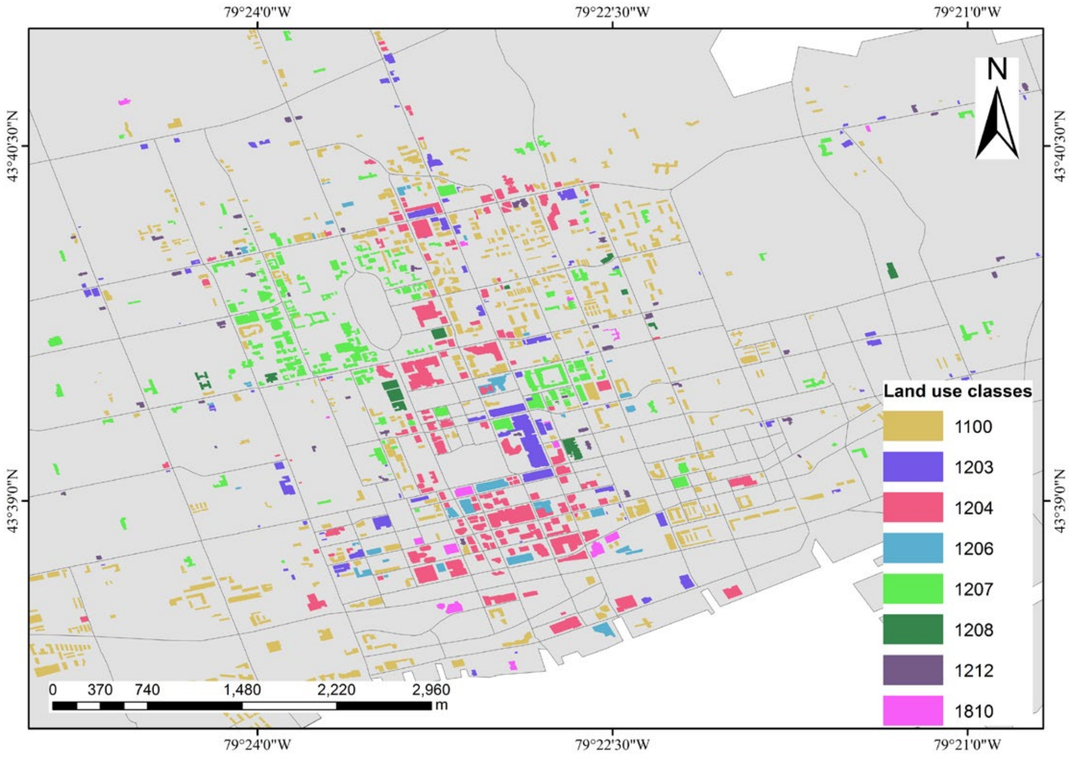

FIGURE 12 Buildings tagged with different land use types

TABLE 3 Validation of mixed land use employing both semantic and dynamic features: Pearson correlations

		Entropy with semantic features	Entropy with labeled buildings
Entropy with semantic features	Pearson correlation	1	0.119
	Sig. (2-tailed)		0.491
	N	36	36
Entropy with labeled buildings	Pearson correlation	0.119	1
	Sig. (2-tailed)	0.491	
	N	36	36
		Entropy with dynamic features	**Entropy with labeled buildings**
Entropy with dynamic features	Pearson correlation	1	0.750**
	Sig. (2-tailed)		0.000
	N	36	36
Entropy with labeled buildings	Pearson correlation	0.750**	1
	Sig. (2-tailed)	0.000	
	N	36	36

**Correlation significant at the 0.01 level (2-tailed).

Anderson (1976), where eight land use types were included in this study, as shown in Table 2. The land use type named "Other Institutional" primarily contains churches and social clubs associated with established organizations. Based on the land use nomenclature, Figure 12 shows part of the classified buildings.

To calculate the diversity of each parcel based on these buildings, the building numbers for each land use type in each parcel are applied to the Shannon entropy for mixed land use evaluation. Here, the parcels without buildings tagged with land use types are not validated for mixed land use. Thus, 36 parcels are used to calculate the Pearson correlation between the Shannon entropy obtained with dynamic features and the labeled buildings. The calculated value of the Pearson coefficient, 0.75 with a p value less than 0.01, indicates a strong positive correlation between the two entropies (as shown in Table 3).

Meanwhile, to quantitively measure the effectiveness of temporal changes in the mixed land use evaluation, a comparative experiment is proposed, which only considers the semantic features in social media. We follow the proposed approaches in Sections 3.3.2 and 3.4 to calculate semantic similarity, and then, we extract the semantic features for mixed land use evaluation. As a result, 14 semantic features are clustered from 20 topics, and then, they are calculated using the Shannon entropy based on the assessment method described in Section 3. To validate the performance of this comparative experiment, the Pearson correlation is calculated and compared with the correlation between entropy with dynamic features and labeled buildings, as shown in Table 3. The Pearson correlation shows a value of 0.119 with p value of 0.491, which shows no significant correlation between the entropy of semantic features and labeled buildings, and this reveals a much worse result compared with the consideration of temporal changes. This shows that temporal changes cannot be ignored in mixed land use evaluation, and this further improves the reliability of the proposed dynamic human activity-driven model.

6 | CONCLUSIONS AND FUTURE WORK

The massive amounts of social media data offered opportunities to evaluate mixed land use in cities from the perspective of human activities. However, there remain challenges derived from two aspects. First, temporal information was ignored in previous research when evaluating mixed land use. Second, the semantic and temporal patterns of different types of land use should be distinguished. In this article, a dynamic human activity-driven model was constructed to evaluate mixed land use using social media data. In terms of the first challenge, this model extracted topics from the textual information of social media data through topic modeling and handled the temporal distributions of topics. Faced with the second challenge, both semantic and temporal similarities were utilized to cluster topics into dynamic features, which were further considered to be indicators for mixed land use evaluation. In the experiments, the Twitter messages were used to evaluate mixed land use in an urban area of Toronto, Canada. A total of 30 temporal topics were extracted through variable textual information, based on 14 dynamic features clustered with the temporal and semantic similarity of topics. These dynamic features were applied to the evaluation of mixed land use using the Shannon entropy. To validate the results from the proposed model, the Shannon entropy calculated from land use type-labeled buildings was utilized to analyze the Pearson correlation with the dynamic features entropy. The Pearson coefficient value of 0.75 largely indicates the efficacy of the proposed model.

Despite a relatively satisfying performance in evaluating mixed land use, some limitations and future work should be considered for the following aspects.

First, as limitations exist in utilizing social media data, including complex data processing and ambiguous textual information, a basic hypothesis that social media data can capture generalized human activities for measuring land use was proposed to cover the above mentioned shortages, as well as representative topics that can be extracted to reduce textual information complexity. Despite the improvements that have been made, these assumptions cannot fully meet the requirements of land use analysis. Additionally, the involvement of multiple sources of crowdsourced data may supplement the characteristics of both human activities and land use. There are opportunities to better measure mixed land use by integrating a variety of features from different data sources.

Second, land use parcels derived from only the main roads were determined to be the basic units of mixed land use evaluation. Compared with regular grids, road-based parcels are capable of correlating regions to a real neighborhood on the ground, and thus show the potential to reveal land use mixture based on the existing urban forms and structure. In fact, mixed land use is sensitive to the spatial scales of selected units, which is known as the modifiable areal unit problem (Openshaw, 1984; Openshaw & Taylor, 1979). Previous studies have analyzed the potential of identifying land use at different scales, including regular grids, communities, buildings, and streets (Liu & Long, 2016; Vanderhaegen & Canters, 2017; Zhu et al., 2017). On the other hand, different stories in buildings may also contain a variety of land uses, bringing more challenges in investigating urban land uses. Temporally, there is a lack of research investigating similar issues, and this reveals the potential to employ the presented methodology to detect the mixed land use variation from a vertical perspective. In particular, since mixed land use is supposed to be quantified in a horizontal (each building) and vertical (each story) scale, the stories of each building will be considered as documents for dynamic feature extraction. Future work will take this issue into account, and explore how to evaluate mixed land use at different spatial scales.

Third, the extracted dynamic features do not correlate to certain land use types. In fact, as mixed land use usually refers to a combination of different uses, the quantification of "mixture" mainly focuses on intensity and diversity, instead of single uses (Yue et al., 2017). Thus, we only took two features as examples to describe land use characteristics and distributions in the experiments (Figure 10). Despite the gap between land use and mixture, limits still remain in further investigation of mixed land use. These include the components of land use types in each parcel and how these land uses change during a specific time period (Tu et al., 2017). Therefore, future studies will focus on decomposing mixed uses and exploring how different land uses could contribute to the urban land use mixture.

Fourth, only the buildings tagged with land use types in OSM were used as reference data to validate the proposed model, because the land use resources are not open to the public and are difficult to obtain. However, the use of such volunteer-based data usually faces challenges, including uncertainty over data completeness and quality (Fonte, Bastin, See, Foody, & Lupia, 2015). Data quality was checked by random sampling and visual interpretation. Land use parcels with sufficient land use type-labeled buildings were selected to avoid data incompleteness. In this case, only 36 parcels were used for validation. This had some effect on the validation of the proposed model. To address this problem, future work will concentrate on developing more effective validation methods and not just calculating the correlation between experimental results and reference data.

Fifth, the proposed approach offers an alternative insight into land use planning. As land use types are strongly correlated with dynamic human activities, this model has the potential to be applied to detecting urban forms and dynamic land use changes, as well as human activity interactions in different land uses. For instance, through modeling the posting messages from social media data to portray different human activities, it can help measure urban vibrancy, which is essential to improve the quality of urban lives (Jacobs, 1961). On the other hand, rather than focusing the proposed approach solely on land use, it would also be interesting for future investigations to utilize the proposed method in other applications, such as natural event detection and urban politics. As attempts have been made in these domains using social media data (Houston et al., 2015; Jin et al., 2017), employing this model with the inclusion of clustering based on semantic similarity will make it possible to extract effective information and monitor target events using social media data.

ACKNOWLEDGMENTS

We would like to thank Prof. Songnian Li for providing the Twitter messages used in this research. Thanks also to the anonymous reviewers for their valuable comments. Hanfa Xing developed the framework and wrote the manuscript. Yuan Meng implemented the method, performed the major part of the experiments, and drafted the manuscript. Yan Shi made substantial contributions to conceptual design and methodological development. This research was supported by the National Natural Science Foundation of China (Grant Nos. 41501420, 41601424), and the Science and Technology Program of Guangzhou, China (Nos.: 201803030034 and 201802030008).

REFERENCES

Abdullahi, S., Pradhan, B., Mansor, S., & Shariff, A. R. M. (2015). GIS-based modeling for the spatial measurement and evaluation of mixed land use development for a compact city. *GIScience & Remote Sensing*, *52*(1), 18–39.

Anderson, J. R. (1976). *A land use and land cover classification system for use with remote sensor data*. Washington, DC: U.S. Geological Survey Professional Paper No. 964.

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, *3*, 993–1022.

Brown, B. B., Yamada, I., Smith, K. R., Zick, C. D., Kowaleski-Jones, L., & Fan, J. X. (2009). Mixed land use and walkability: Variations in land use measures and relationships with BMI, overweight, and obesity. *Health & Place*, *15*(4), 1130–1141.

Cervero, R. (1996). Mixed land-uses and commuting: Evidence from the American Housing Survey. *Transportation Research Part A: Policy and Practice*, *30*(5), 361–377.

Chang, J., Gerrish, S., Wang, C., Boyd-Graber, J. L., & Blei, D. M. (2009). Reading tea leaves: How humans interpret topic models. *In Proceedings of the 22nd International Conference on Advances in Neural Information Processing Systems*. Vancouver, BC, Canada.

Chen, L., Özsu, M. T., & Oria, V. (2005). Robust and fast similarity search for moving object trajectories. In *Proceedings of the 2005 ACM SIGMOD International Conference on Management of Data*. Baltimore, MD: ACM.

Chen, Y., Liu, X., Li, X., Liu, X., Yao, Y., Hu, G., …, Pei, F. (2017). Delineating urban functional areas with building-level social media data: A dynamic time warping (DTW) distance based *k*-medoids method. *Landscape & Urban Planning*, *160*, 48–60.

Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, *1*(2), 224–227.

Fonte, C. C., Bastin, L., See, L., Foody, G., & Lupia, F. (2015). Usability of VGI for validation of land cover maps. *International Journal of Geographical Information Science*, *29*(7), 1269–1291.

Frias-Martinez, V., & Frias-Martinez, E. (2014). Spectral clustering for sensing urban land use using Twitter activity. *Engineering Applications of Artificial Intelligence*, *35*, 237–245.

Gao, S., Janowicz, K., & Couclelis, H. (2017). Extracting urban functional regions from points of interest and human activities on location-based social networks. *Transactions in GIS*, *21*(3), 446–467.

García-Palomares, J. C., Salas-Olmedo, M. H., Moya-Gómez, B., Condeço-Melhorado, A., & Gutiérrez, J. (2018). City dynamics through Twitter: Relationships between land use and spatiotemporal demographics. *Cities*, *72*, 310–319.

Gehrke, S. R., & Clifton, K. J. (2018). An activity-related land use mix construct and its connection to pedestrian travel. *Environment and Planning B*, *45*.

Gervasoni, L., Bosch, M., Fenet, S., & Sturm, P. (2016). A framework for evaluating urban land use mix from crowd-sourcing data. In *Proceedings of the 2016 IEEE International Conference on Big Data*. Washington, DC: IEEE.

Gong, L., Liu, X., Wu, L., & Liu, Y. (2016). Inferring trip purposes and uncovering travel patterns from taxi trajectory data. *Cartography & Geographic Information Science*, *43*(2), 103–114.

Gong, Y., Lin, Y., & Duan, Z. (2017). Exploring the spatiotemporal structure of dynamic urban space using metro smart card records. *Computers, Environment & Urban Systems*, *64*, 169–183.

Grant, J. (2002). Mixed use in theory and practice: Canadian experience with implementing a planning principle. *Journal of the American Planning Association*, *68*(1), 71–84.

Griffiths, T. L., & Steyvers, M. (2004). Finding scientific topics. *Proceedings of the National Academy of Sciences*, *101*(Suppl. 1), 5228–5235.

Heiden, U., Heldens, W., Roessner, S., Segl, K., Esch, T., & Mueller, A. (2012). Urban structure type characterization using hyperspectral remote sensing and height information. *Landscape & Urban Planning*, *105*(4), 361–375.

Houston, J. B., Hawthorne, J., Perreault, M. F., Park, E. H., Goldstein Hode, M., & McElderry, J. A. (2015). Social media and disasters: A functional framework for social media use in disaster planning, response, and research. *Disasters*, *39*(1), 1–22.

Hu, T., Yang, J., Li, X., & Gong, P. (2016). Mapping urban land use by using Landsat images and open social data. *Remote Sensing*, *8*(2), 151.

Hubert, L. (1974). Approximate evaluation techniques for the single-link and complete-link hierarchical clustering procedures. *Journal of the American Statistical Association*, *69*(347), 698–704.

Jacobs, J. (1961). The death and life of great American cities. *Yale Law Journal*, *6*(23), 94–109.

Jin, X., Long, Y., Sun, W., Lu, Y., Yang, X., & Tang, J. (2017). Evaluating cities' vitality and identifying ghost cities in China with emerging geographical data. *Cities*, *63*, 98–109.

Lansley, G., & Longley, P. A. (2016). The geography of Twitter topics in London. *Computers, Environment & Urban Systems*, *58*, 85–96.

Liu, X., He, J., Yao, Y., Zhang, J., Liang, H., Wang, H., & Hong, Y. (2017). Classifying urban land use by integrating remote sensing and social media data. *International Journal of Geographical Information Science*, *31*(8), 1675–1696.

Liu, X., & Long, Y. (2016). Automated identification and characterization of parcels with OpenStreetMap and points of interest. *Environment & Planning B*, *43*(2), 341–360.

Liu, X., Niu, N., Liu, X., Jin, H., Ou, J., Jiao, L., & Liu, Y. (2018). Characterizing mixed-use buildings based on multi-source big data. *International Journal of Geographical Information Science*, *32*(4), 738–756.

Liu, Y., Liu, X., Gao, S., Gong, L., Kang, C., Zhi, Y., & Shi, L. (2015). Social sensing: A new approach to understanding our socioeconomic environments. *Annals of the Association of American Geographers*, *105*(3), 512–530.

Liu, Y., Wang, F., Xiao, Y., & Gao, S. (2012). Urban land uses and traffic 'source-sink areas': Evidence from GPS-enabled taxi data in Shanghai. *Landscape & Urban Planning*, *106*(1), 73–87.

Ma, Y.-S., & Chen, X. (2013). Geographical and statistical analysis on the relationship between land-use mixture and home-based trip making and more: Case of Richmond, Virginia. *Journal of Urban and Regional Analysis*, *5*(1), 5.

Marteau, P.-F. (2009). Time warp edit distance with stiffness adjustment for time series matching. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, *31*(2), 306–318.

Niu, N., Liu, X., Jin, H., Ye, X., Liu, Y., Li, X., & Li, S. (2017). Integrating multi-source big data to infer building functions. *International Journal of Geographical Information Science*, *31*(9), 1871–1890.

Openshaw, S. (1984). *The modifiable areal unit problem: Concepts and techniques in modern geography*. Norwich: Geo Books.

Openshaw, S., & Taylor, P. J. (1979). A million or so correlated coefficients: Three experiments on the modifiable areal unit problem. In N. Wrigley (Ed.), *Statistical applications in the spatial sciences* (pp. 127–144). London: Pion.

Ríos, S. A., & Muñoz, R. (2017). Land use detection with cell phone data using topic models: Case Santiago, Chile. *Computers, Environment & Urban Systems*, *61*, 39–48.

Seasons, M. (2014). Does mixed-use development benefit everyone? Housing affordability in a changing labour market (Unpublished M.A. Thesis). Waterloo, ON, Canada: University of Waterloo.

Serra, J., & Arcos, J. L. (2014). An empirical evaluation of similarity measures for time series classification. *Knowledge-Based Systems*, *67*, 305–314.

Song, Y., & Knaap, G.-J. (2004). Measuring the effects of mixed land uses on housing values. *Regional Science & Urban Economics*, *34*(6), 663–680.

Song, Y., Merlin, L., & Rodriguez, D. (2013). Comparing measures of urban land use mix. *Computers, Environment & Urban Systems*, *42*, 1–13.

Tormene, P., Giorgino, T., Quaglini, S., & Stefanelli, M. (2009). Matching incomplete time series with dynamic time warping: An algorithm and an application to post-stroke rehabilitation. *Artificial Intelligence in Medicine*, *45*(1), 11–34.

Tu, W., Cao, J., Yue, Y., Shaw, S.-L., Zhou, M., Wang, Z., & Li, Q. (2017). Coupling mobile phone and social media data: A new approach to understanding urban functions and diurnal patterns. *International Journal of Geographical Information Science*, *31*(12), 2331–2358.

Vanderhaegen, S., & Canters, F. (2017). Mapping urban form and function at city block level using spatial metrics. *Landscape & Urban Planning*, *167*, 399–409.

Wang, X., Mueen, A., Ding, H., Trajcevski, G., Scheuermann, P., & Keogh, E. (2013). Experimental comparison of representation methods and distance measures for time series data. *Data Mining & Knowledge Discovery*, *26*(2), 275–309.

Williams, W., Lance, G., Dale, M. B., & Clifford, H. (1971). Controversy concerning the criteria for taxonometric strategies. *Computer Journal*, *14*(2), 162–165.

Xing, H., Meng, Y., Hou, D., Song, J., & Xu, H. (2017). Employing crowdsourced geographic information to classify land cover with spatial clustering and topic model. *Remote Sensing*, *9*(6), 602.

Yao, Y., Li, X., Liu, X., Liu, P., Liang, Z., Zhang, J., & Mai, K. (2017). Sensing spatial distribution of urban land use by integrating points-of-interest and Google Word2Vec model. *International Journal of Geographical Information Science*, *31*(4), 825–848.

Yao, Y., Liu, X., Li, X., Zhang, J., Liang, Z., Mai, K., & Zhang, Y. (2017). Mapping fine-scale population distributions at the building level by integrating multisource geospatial big data. *International Journal of Geographical Information Science*, *31*(6), 1220–1244.

Yuan, J., Zheng, Y., & Xie, X. (2012). Discovering regions of different functions in a city using human mobility and POIs. In Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Beijing: ACM.

Yue, Y., Zhuang, Y., Yeh, A. G., Xie, J.-Y., Ma, C.-L., & Li, Q.-Q. (2017). Measurements of POI-based mixed use and their relationships with neighbourhood vibrancy. *International Journal of Geographical Information Science*, *31*(4), 658–675.

Zhu, D., Wang, N., Wu, L., & Liu, Y. (2017). Street as a big geo-data assembly and analysis unit in urban studies: A case study using Beijing taxi data. *Applied Geography*, *86*, 152–164.

How to cite this article: Xing H, Meng Y, Shi Y. A dynamic human activity-driven model for mixed land use evaluation using social media data. *Transactions in GIS*. 2018;00:1–22. https://doi.org/10.1111/tgis.12447